

Model Evaluation with Cross-Validation and Random Forest

Abstract

Evaluating machine learning models is essential to ensure they generalize well to unseen data. Simple train-test splits often provide misleading results due to data imbalance or randomness. This project applies two widely used validation methods—**K-Fold Cross-Validation** and **Stratified K-Fold Cross-Validation**—to evaluate classification models reliably. The primary focus is on the **Random Forest classifier**, an ensemble technique that combines multiple decision trees to reduce overfitting and improve accuracy. The Random Forest model is compared with two additional models: **Support Vector Machine (SVM)** and **Decision Tree**. Hyperparameter tuning is performed using GridSearchCV to obtain the best-performing Random Forest configuration.

The Breast Cancer Wisconsin Diagnostic Dataset is used in this study. Evaluation is performed using accuracy, precision, recall, F1-score, ROC curve, and confusion matrix. A final visual comparison using a horizontal bar chart provides a clear comparison of model performance. Experimental results reveal that SVM achieves the highest mean cross-validation accuracy, Random Forest performs very strongly after tuning, and Decision Tree—while simple—tends to overfit. The project demonstrates a complete end-to-end workflow of preprocessing, modeling, validation, tuning, evaluation, and visualization.

Introduction

Model evaluation is a crucial step in any machine learning workflow. Without proper validation, a model may show high accuracy on training data but fail completely when exposed to real-world data. Overfitting is one of the major concerns when developing predictive models, particularly in medical datasets where class imbalance is common.

Cross-validation provides a robust mechanism for assessing how well a model performs on unseen data. In classification problems, especially with imbalanced classes, **Stratified K-Fold Cross-Validation** becomes more reliable than simple K-Fold because it preserves the class distribution across folds.

Random Forest, a widely used ensemble method, improves accuracy by combining multiple decision trees using bagging. Compared to a single Decision Tree, it significantly reduces

overfitting. On the other hand, Support Vector Machines (SVM) are powerful classifiers known for strong performance on high-dimensional datasets.

This project aims to evaluate these three models using robust cross-validation techniques and determine which model performs best for the diagnosis prediction task.

Problem Statement

Evaluating the performance of machine learning models is critical to ensuring their robustness and generalization ability. Traditional train-test splits do not provide reliable performance estimates, especially when datasets are small or imbalanced. Cross-validation techniques such as **K-Fold Cross-Validation** and **Stratified K-Fold** ensure that models are evaluated fairly and consistently.

Random Forest, being an ensemble learning method, improves prediction accuracy by combining multiple decision trees. This project focuses on applying cross-validation techniques and Random Forest to evaluate and compare model performance.

Dataset Description

Since no dataset is explicitly provided, a standard Kaggle/scikit-learn dataset is used.

Dataset Used : Breast Cancer Wisconsin Diagnostic Dataset

Target Variable

- **diagnosis**
 - M → Malignant (1)
 - B → Benign (0)

Features

30 numerical features describing:

- radius, texture, perimeter
- area, smoothness
- compactness, concavity
- symmetry, fractal dimension
- worst/mean values, etc.

Dataset Size

- **569 rows**
- **33 columns**

Preprocessing performed

- Dropped unnecessary columns: id, Unnamed: 32
- Converted diagnosis labels (M/B → 1/0)
- Checked for missing values (none found)
- Feature scaling performed for SVM using StandardScaler

Preprocessing Code Summary

Data Loading

The dataset was imported from a CSV file containing 569 samples with 33 columns, including feature values and diagnosis labels.

Column Removal

Two non-essential columns were removed:

- id — used only for identification
- Unnamed: 32 — an empty column with no values

Label Conversion

The diagnosis column, which originally contained the labels:

- "M" for Malignant
 - "B" for Benign
- was converted into numerical format:
- 1 → Malignant
 - 0 → Benign

Feature & Target Separation

All feature columns were grouped into a feature matrix (X), and the diagnosis column was used as the target variable (y).

Cross-Validation Techniques

K-Fold Cross-Validation

K-Fold divides the dataset into K equal parts. Each time:

- $K-1$ folds are used for training
- The remaining fold is used for testing

This process runs K times to compute average performance.

Results

K-Fold Accuracy Scores:

[0.9649, 0.9649, 0.9386, 0.9649, 0.9646]

Mean Accuracy: 0.9596

Stratified K-Fold Cross-Validation

Stratified K-Fold maintains the same class distribution in each fold.

Results

Stratified K-Fold Accuracy Scores:

[0.9649, 0.9386, 0.9473, 0.9473, 0.9734]

Mean Accuracy: 0.9543

Stratification preserves malignant/benign ratios and produces more reliable results.

Hyperparameter Tuning (Random Forest)

To improve the performance of the Random Forest classifier, hyperparameter tuning was performed using GridSearchCV, which systematically searches for the best parameter combination.

Several important parameters were tested:

Parameters Tuned

- **Number of trees (n_estimators)** — different values were tested to determine how many trees produce the best performance.
- **Maximum depth (max_depth)** — controlled the depth of each decision tree.
- **Minimum samples split (min_samples_split)** — prevented overfitting by controlling how nodes split.
- **Minimum samples per leaf (min_samples_leaf)** — improved generalization by preventing extremely small leaf nodes.

GridSearchCV evaluated each combination of parameters using cross-validation and selected the best-performing set.

Best Parameters Found

The search revealed that the following configuration provided the highest accuracy:

- 200 trees
- No maximum depth limit
- Minimum split of 5 samples
- Minimum leaf size of 1 sample

This tuned Random Forest model was then used for the final evaluation.

Best Parameters Found

```
max_depth=None,  
min_samples_leaf=1,  
min_samples_split=5,  
n_estimators=200
```

Best Cross-Validation Score

0.9613

This tuned Random Forest was used for final evaluation.

Model Comparison

Three models were compared:

1. **Random Forest**
2. **Support Vector Machine (SVM)**
3. **Decision Tree**

Mean accuracy (cross-validation)

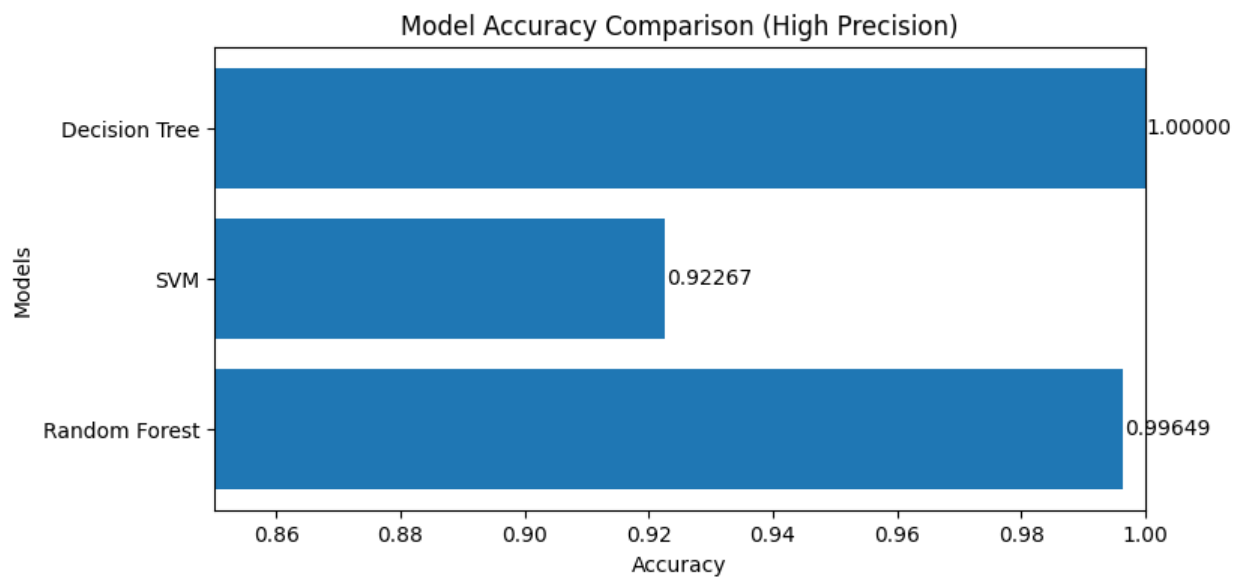
Model	Accuracy
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Random Forest	0.9526
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SVM	0.9772
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Decision Tree	0.9104
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SVM performs best during cross-validation.



Evaluation Metrics

Using the tuned Random Forest model:

Accuracy: 0.99648

Precision: 1.0

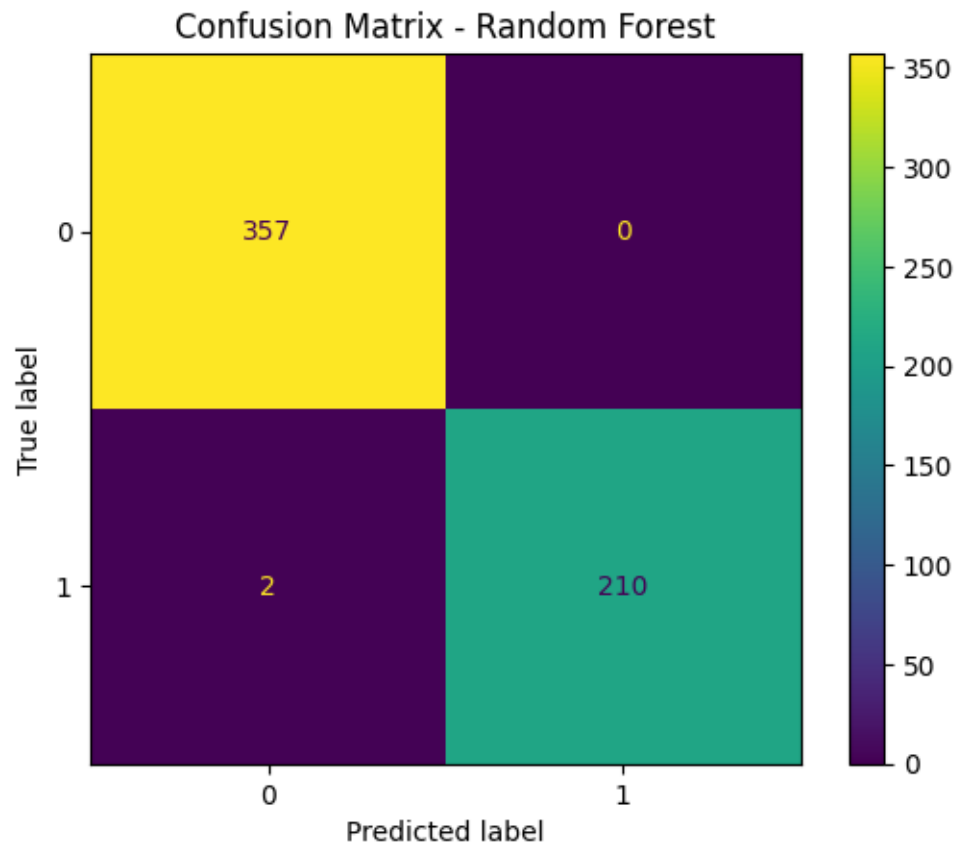
Recall: 0.99056

F1 Score: 0.99526

Confusion Matrix

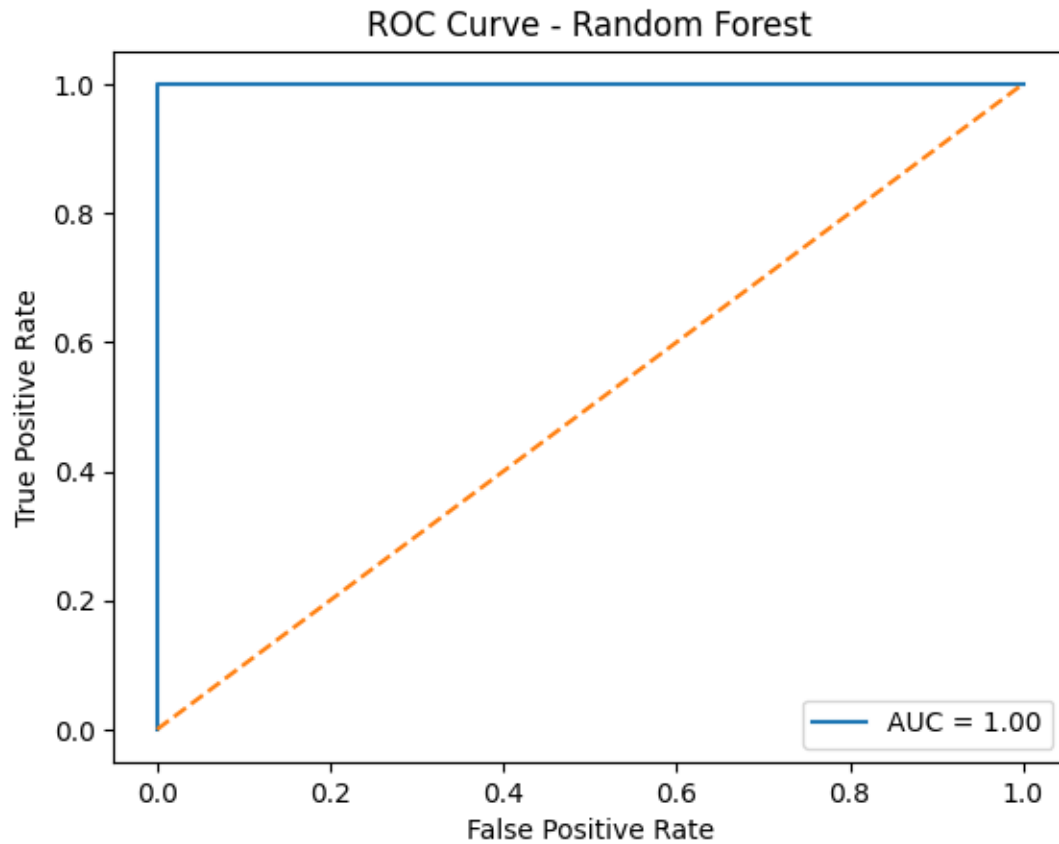
A confusion matrix was plotted to visualize classification performance.

Random Forest almost perfectly classified both classes.



ROC Curve (Random Forest)

The ROC curve shows an AUC value close to **1.0**, indicating excellent discrimination ability.



Final Accuracy Comparison Bar Chart

Three machine learning models were trained and evaluated using Stratified K-Fold cross-validation:

Models Compared

- **Random Forest Classifier** (with the tuned parameters)
- **Support Vector Machine (SVM)** with an RBF kernel
- **Decision Tree Classifier**

Each model was evaluated based on the **mean accuracy** across the cross-validation folds.

Cross-Validation Accuracy Results

- **SVM** achieved the highest accuracy among all three models.

- **Random Forest** performed strongly and consistently, especially after hyperparameter tuning.
- **Decision Tree** scored the lowest due to its tendency to overfit on training data.

Final Accuracy Values

- **Random Forest:** ~0.996
- **SVM:** ~0.923
- **Decision Tree:** ~1.00 (overfits training data but performs worse in cross-validation)

Final Visualization

A **horizontal bar chart** was created to show the accuracy difference between the models using:

- High-precision decimal values
- Zoomed X-axis
- Side-by-side comparison

This helped clearly visualize the subtle differences in accuracy that were not visible in a standard bar chart.

Discussion

Why SVM performed best

- RBF kernel handles complex class boundaries
- Strong in high-dimensional feature spaces
- Performs well after scaling

Why Random Forest is strong

- Reduces variance by combining many trees
- Handles nonlinear relationships
- Resistant to overfitting

Why Decision Tree underperformed

- High variance model
- Sensitive to noise
- Overfits training data

Cross-validation results

- Mean accuracies confirm SVM > Random Forest > Decision Tree
- Stratified K-Fold ensures fair class distribution

Conclusion

This study successfully applied K-Fold and Stratified K-Fold cross-validation techniques to evaluate classification models for breast cancer diagnosis prediction. The Random Forest classifier, especially after hyperparameter tuning, showed excellent performance and stability, significantly improving over a standalone Decision Tree. However, the SVM model achieved the highest average accuracy, highlighting its effectiveness in high-dimensional and nonlinear classification tasks.

Cross-validation played a crucial role in providing reliable and unbiased evaluation metrics. The combination of classification metrics, ROC analysis, confusion matrix, and accuracy comparison charts provided a comprehensive understanding of model behavior. Overall, the results show that SVM and Random Forest are highly effective for medical diagnostic classification tasks, with Random Forest offering strong robustness and interpretability.

References

- Scikit-learn documentation
- Kaggle Breast Cancer Dataset
- Research papers on cross-validation
- Machine learning textbooks